Adaptive MultiScale Segmentations for Hyperspectral Image Classification

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***Abstract*— The number of superpixels (i.e., segmentation scale) is crucial for spectral–spatial hyperspectral image (HSI) classification. Existing methods always set the segmentation scale through a manually experimental strategy, which is time-consuming and unsuitable for various complicated practical applications. The information fusion of complementary multiple scales is proven to be more effective than the single scale for HSI classification, but the scale level is still set manually. In this article, we propose a novel adaptive multiscale segmentations (AMSs) method that can automatically provide a set of suitable scales that are adapted to different hyperspectral data. Specifically, based on the assumption that the segmentation scale of HSI is related to the image complexity itself, the texture ratio and the number of land cover classes are used to examine a candidate scale pool. A good scale means that it contains a small spectral difference between pixels within the same superpixel (intrasuperpixel discrimination index) and a large discrepancy between neighboring superpixels (intersuperpixel discrimination index). Thus, an intra–interscale discrimination index is defined and applied to depict the characteristics of the scale. Then, the scale with the best intra–inter discrimination index, which usually has satisfactory performance, is treated as the initially selected scale. The remaining suitable scales are iteratively compared with the selected ones and then added to the target scale pool, until the newly added scale can no longer provide significantly complementary information. Extensive experimental results on three HSI data sets have demonstrated the effectiveness of the proposed AMS when compared with state-of-the-art methods.**

***Index Terms*— Complementary information, hyperspectral image (HSI) classification, intersuperpixel discrimination index, intrasuperpixel discrimination index, multiscale segmentations.**

I. Introduction

HYPERSPECTRAL images (HSIs) can capture numerous and continuous spectral bands of land cover/land use, and they have been promoted for various remote sensing applications, such as smart city [1], mineralogy [2], ecological monitoring [3], national defense [4], and so on. As one of the major tasks in the HSI processing community, classification has drawn great and increasing attention in the past decades [5]. Early studies in the HSI classification literature are generally concerned with the pixel-wise spectral features of hyperspectral imagery [6], [7]. Although pixel-wise approaches can be easily implemented [8], their performances are still far from satisfactory due to limitations of high dimensionality and spectral variabilities [9], [10]. HSI is spatially dependent and contains clustered pixels. The spatial constraint can provide remarkable additional information for improving pixel-wise classification. Therefore, spectral–spatial approaches are significantly promoted and have become a mainstream topic in HSI classification [11], [12].

With regard to obtaining the spatial information, most approaches can be categorized into two types [13]: a system based on the crisp neighborhood [14], [15] and a system based on the adaptive neighborhood [16], [17]. Compared with the former, the latter exploits flexible spatial information by using shape-adaptive regions, which is more widely used in spectral–spatial HSI classification. In the adaptive neighborhood system, superpixel segmentation is the most common method for generating homogeneous regions [18]–[20]. The number of superpixel regions (i.e., segmentation scale) will affect the discrimination of spectral–spatial features, and how to select a reasonable number of superpixels is always a very important issue. However, the labels of test data are unknown in practice, and determining the optimal scales without the groundtruth is impossible. The classes and distribution of land cover always differ on diverse HSIs, which indicate that fixed scales are difficult to adapt to various hyperspectral scenarios. For reasons that texture information can be used to measure the complexity of HSIs to some extent, some researchers attempt to directly calculate the number of scale S^f by exploiting the texture ratio R of HSI [21], [22], such that $S^f = S^{\text{base}} * R$, where S^{base} is a basic segmentation number. R is driven by hyperspectral data, and the obtained S^f seems good and sometimes approximates the best scale. However, S^{base} is predefined experientially, which implies that the setting of S^f is still manually dependent.

In order to cope with the manual dependence dilemma of the single segmentation scale, some studies utilize a multiscale fusion framework for HSI classification [21], [23]–[25]. The underlying assumptions of this framework manifest in two aspects. On the one hand, HSI in different scales shares the same pixels, and complementary spatial information naturally

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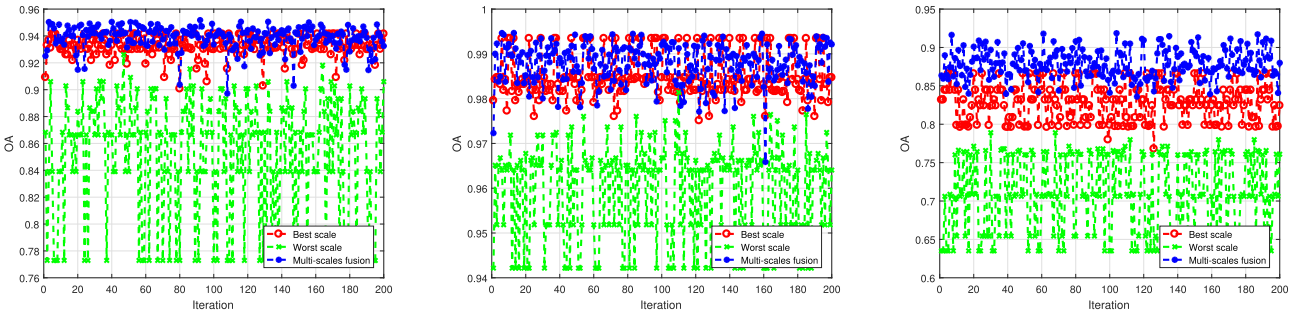


Fig. 1. OAs of the multiscale fusion over SVM for 200 iterations on three standard hyperspectral data sets, for each iteration we randomly choose five different scales in the interval of [10 : 10 : 200] and then fuse the classification results of these scales. For each group of five scales, the results of the best single scale, worst single scale, and multiscale fusion are labeled in red "o," green "x," and blue "*" colors, respectively. As can be seen, the blue points are sometimes above the red ones, especially on the Salinas data set. (a) Indian Pines data set. (b) Salinas data set. (c) University of Pavia data set.

exists between multiple scales [26]–[29]. On the other hand, elaborating with multiple scales is better than that with a single scale. Specifically, in a general framework, the multiscale superpixel regions of the original HSI are first generated on the basis of a given group of scales. Then, the spectral–spatial features of different scales are extracted by a certain integration strategy. Finally, the classification maps of all scales are obtained and subsequently fused to produce the eventual result. However, the multiple scales in the existing works are still manually set, which further result in two unavoidable problems. First, a universally setting scheme of multiple scales is absent, which hinders the matching of various HSIs. Second, introducing more scales are not necessarily better, the fusion result of all scales in a much larger interval is not only time-consuming but also probably unsatisfactory, and the selection of scales becomes crucial instead of casual. We demonstrate the point by the classification results shown in Fig. 1 on three typical HSI data sets, i.e., Indian Pines, Salinas, and University of Pavia. These experiments are conducted based on multiscale fusion for 200 iterations; for each iteration, we randomly choose five different scales in the interval of [10:10:200], and then fuse the classification results of these scales. The classification model adopts support vector machines (SVMs) [6], and the evaluation metric is overall accuracy (OA). Here, although the OAs after multiscale fusion (blue lines) are more likely to be higher on the University of Pavia, the performance based on the best of five scales is sometimes better, especially on the Salinas. The effects of different fusions vary greatly.

In the literature of superpixel segmentation [30]–[32], the major principle is maintaining the homogeneity of each superpixel so as to obtain clear boundaries between superpixels. In other words, the homogeneity between pixels within the same superpixel and the inhomogeneity between neighboring superpixels can be used to describe the characteristic of a scale and then evaluate the similarity between scales. Therefore, following the multiscale fusion framework, this article proposes a novel adaptive multiscale segmentations (AMSs) method for HSI classification by exploiting the homogeneity and inhomogeneity of superpixels. As shown in Fig. 2, the AMS method mainly includes three steps.

1) *Candidate Scale Pool Definition:* The segmentation scale of HSI is related to the complexity of texture information and land cover classes [21], [22]. Thus, without setting S^{base}, we can define the candidate scale pool by comprehensively examining the texture ratio R and the number of classes, which are driven by hyperspectral data. Then, candidate scales are generated by entropy rate superpixel segmentation (ESR) [33].
2) *Scale Discrimination Index Calculation:* A good scale should have a smaller spectral difference between pixels within the same superpixel (intrasuperpixel discrimination index) and a larger difference between neighboring superpixels (intersuperpixel discrimination index) compared with a bad scale, then an intra–interscale discrimination index can be measured.
3) *Adaptive Multiple Scales Determination:* The scale with the best discrimination index, which often leads to good classification performance, is regarded as the initially selected scale. Then, suitable scales, which are compared with the selected ones, are iteratively chosen from the candidate scale pool, until complementary information can no longer be evidently provided by the newly added

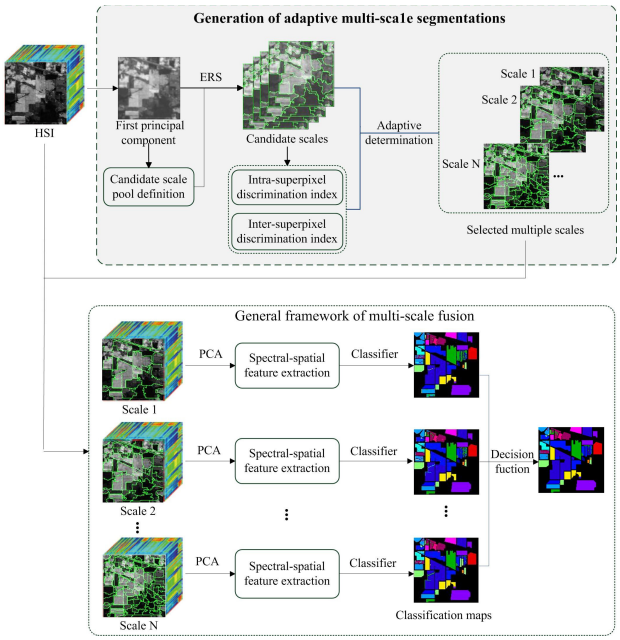


Fig. 2. Flowchart of the proposed AMS method for HSI classification based on multiscale fusion.

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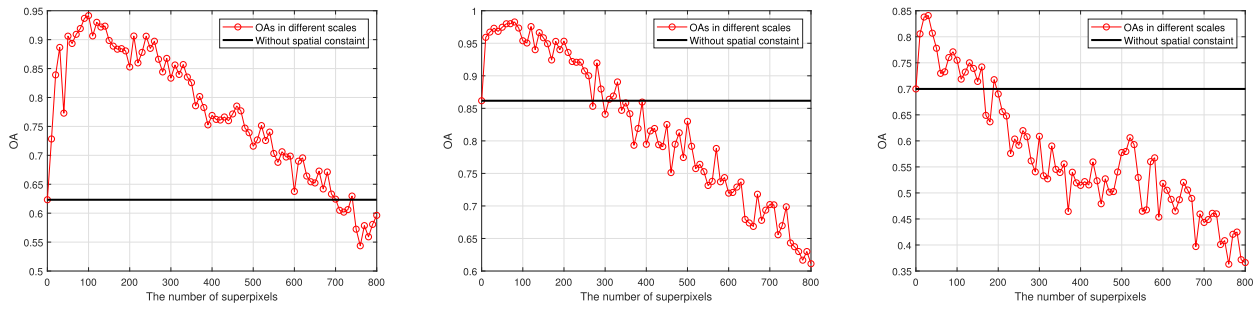


Fig. 3. OAs of scales in the interval of [0 : 10 : 800] on three HSI data sets. Notably, no spatial constraint exists when the number of superpixels is 0, which is labeled in black line. (a) Indian Pines data set. (b) Salinas data set. (c) University of Pavia data set.

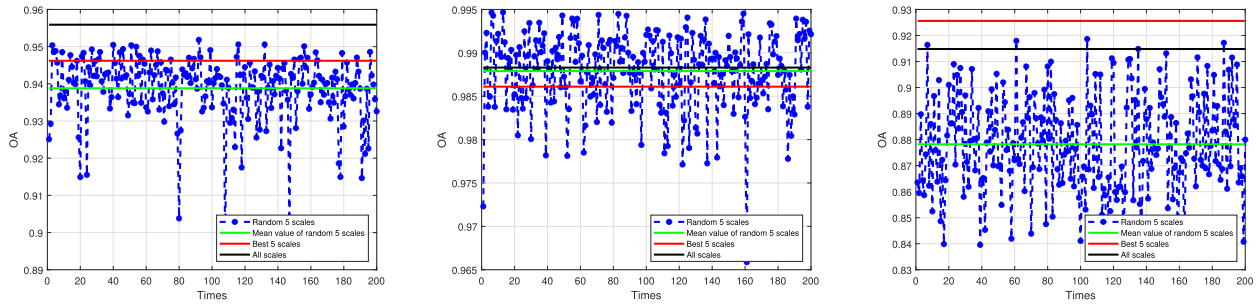


Fig. 4. Results of comparison of the fusions of the random five scales and best five scales in the interval of [10 : 10 : 200], all scales in the interval of [10 : 200] are labeled in blue "*," red line, and black line, respectively. The mean value of the random five scales is also marked in green line for better presentation. (a) Indian Pines data set. (b) Salinas data set. (c) University of Pavia data set.

scales. Finally, those determined scales are used for multiscale fusion, and the eventual classification result can be obtained by a voting strategy [34].

The main contributions of this article are as follows.

1) A novel intra–interscale discrimination index is presented to measure the similarity between scales. Findings suggest that the index is positively related to classification performance to a certain degree, and it can be used to automatically seek an appropriate single scale.
2) Compared with the current manual strategies, a simple yet effective AMS method without experimental parameters is proposed to adaptively determine the suitable scales for multiscale HSI classification. Extensive experimental results compared with segmentation-based baselines and state of the art have demonstrated the robustness of the proposed method.

The rest of this article is organized as follows. Section II presents the analysis of the impact of segmentation scale on HSI classification. Section III presents in detail the methodology of the proposed AMS. Experimental results and discussions regarding the comparison of the proposed method with the baselines are given in Section IV. Section V presents the conclusion.

II. Influence of Segmentation Scale on HSI Classification

Many studies have mentioned that the selection of the segmentation scale is not trivial, but it can evidently affect the performance of spectral–spatial classification [35]–[38]. In this section, we further quantitatively demonstrate the impact of the segmentation scale, which is still absent in existing studies. The demonstration is concretely conducted in three steps. First, the OA of each scale in a large interval of [0 : 10 : 800] is summarized, which indicates that the performance trend along with the increase of scales can be observed. Second, in an interval of [10 : 10 : 200], the fusion result of the scales with top-n performance is compared with that of the random five scales, where n is set to 5 for equal comparison. Third, the fusion result of all scales in the interval of [10 : 200] is evaluated to test whether the multiscale fusion can definitely benefit from using more scales. The settings of the data sets and the classification model are similar to those in Fig. 1. Three observations can be drawn on the basis of the results.

1) The OA curve shown in Fig. 3 has a short rise and a long fall trends with the growth of the scale, and the selection of the scale has been confirmed crucial for HSI classification. Moreover, each data set has a best scale, but no universally fixed scale can be adapted for all three data sets. The performances in the case where without spatial constraint (the number of superpixels is set to 0) is compared. Appropriate scales can significantly improve the classification results, whereas the inappropriate scales can lead to the sharp decline of OAs, especially when the number of superpixels is further increased.
2) In Fig. 4, the fusion results of the random five scales are used to simulate the varying selected five scales. The differences between the top and bottom OAs are much

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large at 5.46%, 2.88%, and 8.26% on Indian Pines, Salinas, and PaviaU, respectively. Thus, the unreasonable groups of multiple scales will probably result in poor performance, and the classification results are sensitive to the selection of scales.

3) The fusion results of the random five scales are compared with those of the best five scales. The results are often suboptimal and even poorer than the mean result on the Salinas data set, demonstrating that the best scales are not always the optimal choice for multiscale fusion. Moreover, the results of all scales are poor on the Salinas data set and worse than that of the best five scales on the University of Pavia. Thus, the classification results cannot definitely benefit from using more scales. By comprehensively considering the presentation in Figs. 1 and 4, the fusion results of the scales with ordinary classification performance can also be effective and sometimes even better than those of the best scales and all scales.

III. Proposed Method

This section gives a detailed explanation of the proposed AMS method. Our core idea is to adaptively seek a set of scales with complementary information, which is proven important for multiscale fusion in the HSI classification. The flowchart in Fig. 2 includes three stages. First, the candidate scale pool is defined by exploiting the number of classes and the texture complexity. Second, an intra–interscale discrimination index is employed to measure the similarity between scales. Third, suitable scales are iteratively selected and then added to the determined set until the newly added ones can no longer provide obvious complementary information.

A. Definition of Candidate Scale Pool

The candidate scale pool is necessary and important in selecting a group of suitable scales. The candidates with a very small-scale interval probably have little complementarity, whereas those in a much larger-scale interval may have many improper scales with extremely poor performance, such as the examples far below the black lines in Fig. 3.

In single-scale spectral–spatial HSI classification, texture complexity is related to the fragmentation of HSI to some extent, and it is commonly used to determine a suitable scale [21]. Specifically, the edge image of HSI is generated by certain texture detectors, e.g., Canny filters [39], Laplacian of Gaussian (LoG) operator, and Sobel operator [40]. Then, the number of edge pixels N^{edge} is compared with the whole number of hyperspectral pixels N^{base} to compute the texture ratio R. Finally, the determined scale S^f is calculated by the product of R and a basic scale S^{base}. However, S^f cannot be directly applied to AMS. Considering that S^{base} is set experimentally and may conflict with our adaptive strategy, an available scale pool is required rather than a single scale for multiscale fusion. In dealing with the problem, the number of land cover classes and the R that are both hyperspectrally driven by data, are comprehensively exploited to generate the candidate scale pool, i.e., the lower and the upper boundaries of the candidates interval.

1) Lower Boundary: In the ideal case, all pixels of the same class are gathered together in the same homogeneous region. The number of classes C of a HSI I can match the number of superpixels, which is then used to define the lower boundary of candidates interval, S^{lower}.

2) Upper Boundary: The distribution of surface objects is so complicated in practice that hyperspectral pixels with the same classes are often scattered rather than clustered, especially when the number of classes is large. In determining the upper boundary, the larger the class number is, the more the superpixels that will naturally exist. However, if the pixels with the same classes are more likely to be dispersed, then elaborate and complicated segmentation is probably ill-suited to obtain a robust spectral–spatial constraint. For example, the fragmentation of the University of Pavia is higher than those of Indian Pines and Salinas, but its optimal value for the optimal single scale is the smallest, as shown in Fig. 3. Therefore, the texture ratio R of the first principal component [41] of HSI can be detected based on the Sobel operator, and the upper boundary of candidates interval is $S^{\text{upper}} = C/R$.

The candidate scale pool can be defined by $[S^{\text{lower}} \quad S^{\text{upper}}]$, that is

$$\text{Interval}(S) = [C \quad C/R]. \tag{1}$$

In general, C is always regarded as a known constant in the HSI classification task. But in practice, some pixels of unknown categories may exist in the HSI. The real number of land cover categories, $\overline{C}$, can be bigger than C, then scale pool $\overline{S}$ referred to $\overline{C}$ is also larger than S. Fortunately, as shown in Figs. 3 and 5, the OAs tend to be lower along with the increase of the scale especially when the scale is oversize, and scales with the optimal classification performance and intra–inter discrimination indexes almost locate in a range of small scales. Therefore, S can also be available even in the situation of some unknown categories.

B. Calculation of Scale Discrimination Index

In selecting a group of suitable scales for the candidates, the core issue is to measure the similarity between scales. Generally, in the case of good scales, the spectral difference between pixels within the same superpixels, i.e., intrasuperpixel discrimination index, is smaller than that in the case of bad scales. Nevertheless, compared with small scales, large scales will generate more superpixels with less yet same-class pixels. The intrasuperpixel discrimination index probably decreases with the growth of scales, whereas much larger scales can lead to poor performance due to the shortage of available spatial constraints. Thus, the intersuperpixel discrimination index is additionally introduced to describe the discrepancy between neighboring superpixels. A high intersuperpixel discrimination index indicates pixels from diverse superpixels that seem to belong to different classes. In summary, the similarity between scales can be acquired by measuring the intra–interscale discrimination indexes, which are the combination of intrasuperpixel and intersuperpixel.

1) IntraSuperpixel Discrimination Index: Assuming that a set of candidate scales, $S = \{s_1, s_2, \ldots, s_N\}$, $S \subseteq [C : k : C/R]$, k is set to 5 for all the experiments to reduce

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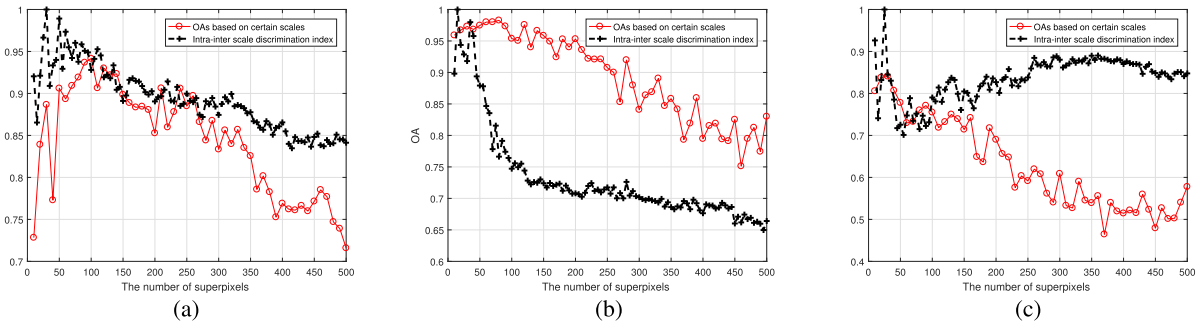


Fig. 5. OA performance and normalized intra–interscale discrimination index according to the number of superpixels. (a) Indian Pines data set. (b) Salinas data set. (c) University of Pavia data set.

the computation complexity. The ith scale has M superpixel regions, $s_i = \{s_{i,1}, s_{i,2}, \ldots, s_{i,M}\}$. Then, the intrasuperpixel discrimination index of s_i is

$$f_{\text{intra}}(s_i) = \frac{\sum_{j=1}^{M} \sigma_{s_{i,j}}}{M} \tag{2}$$

where $\sigma_{s_{i,j}}$ is the variance of the jth superpixels of s_i.

2) InterSuperpixel Discrimination Index: Superpixel segmentation aims to distinguish the edge pixels of adjacent homogeneous regions rather than those of any pair regions, which indicates that the spectral difference between neighboring shape-adaptive regions can be adopted. Let the neighbors of $s_{i,j}$ be represented by $s_{i,j}^{\upsilon}$, $\upsilon = 1, 2, \ldots, P$, and the intersuperpixel discrimination index of $s_{i,j}$, is denoted as

$$f_{\text{inter}}(s_{i,j}) = \frac{\sum_{\upsilon=1}^{P} \text{dist}\left(\text{mean}(s_{i,j}), \quad \text{mean}\left(s_{i,j}^{\upsilon}\right)\right)}{P} \tag{3}$$

where mean$(\cdot)$ is the average value of the pixels in the corresponding superpixel regions, and dist$(\cdot)$ is a cosine distance. Then, the overall intersuperpixel discrimination index of s_i is

$$f_{\text{inter}}(s_i) = \frac{\sum_{j=1}^{M} f_{\text{inter}}(s_{i,j})}{M}. \tag{4}$$

By considering that good scales probably have a small intrasuperpixel discrimination index and a large intersuperpixel discrimination index, the final intra–interscale discrimination index can be represented by a fusion of (2) and (4) as follows:

$$f(s_i) = \exp(-f_{\text{intra}}(s_i)) + f_{\text{inter}}(s_i) \tag{5}$$

where $f_{\text{intra}}(s_i)$ and $f_{\text{inter}}(s_i)$ are normalized before the fusion process.

The OAs of scales and the corresponding $f(s_i)$ are tested and contrasted on three HSI data sets to demonstrate the availability of the intra–interscale discrimination index. As shown in Fig. 5, the two observations are as follows. First, a short upward then a long downward trend exists similarly on the OAs and the normalized scale discrimination indexes, especially for the Indian Pines and Salinas data sets. This finding means that $f(s_i)$ is positively related to the classification result to some degree. Second, although the scale with the maximum of $f(s_i)$ is not the optimal one, it is still close to the best single scale and has good performance; that is to say, $f(s_i)$ can also be used to adaptively choose a suitable single scale. Therefore, although the intra–interscale discrimination index has no strict correspondence to the classification result, it can still feasibly depict the characteristic of scales and can be availably applied to HSI classification.

C. Determination of Adaptive Multiple Scales

An initial scale is needed to seek the complementary scales. Scale with good performance is a better choice compared with random initialization. On basis of the discussion in Section III-B, the intra–inter discrimination index has a positive correlation with OA to some extent. In particular, we can select a suitable scale with maximal value of $f(s_i)$, which is treated as the initial scale s_0 in the candidate scale pool S

$$s_0 = \arg\max_{s_i} f(s_i) \quad \forall s_i \in S. \tag{6}$$

The complementary information between scales is important for effective multiscale fusion [42], [43]. On the one end, current manual strategies also focus on selecting much different scales in large-scale intervals. Meanwhile, the scales with the best classification performance are always considered to be in a neighboring interval as shown in Fig. 3, but the fusion results of those scales are often suboptimal and even sometimes poor in Fig. 4. Hence, following the principle of enhancing the complementarity, the suitable scales are iteratively chosen until no significantly complementary information can be added. Specifically, let $S^t = \{s^1, s^2, \ldots, s^t\}$ be a set of selected scales at time t, $s^1 = s_0$. The set at time $t+1$ is denoted as

$$S^{t+1} = \left\{ S^t, \arg\max_{s_i} \frac{\sum_{s_l \in S^t} |f(s_i) - f(s_l)|}{t} \right\} \quad \forall s_i \in S \;\&\; s_i \notin S^t. \tag{7}$$

Complementary information is a relative concept and regarded as the difference between additional information of s^t and s^{t-1} in this article. Therefore, we first calculate the additional information of s^t, which is the average difference between s^t and previously selected scales

$$y(s^t) = \text{mean}\left(\sum_{s_l \in S^{t-1}} |f(s^t) - f(s_l)| \right). \tag{8}$$

Then, the complementary information provided by s^t, namely comp(s^t) is defined by the difference between $y(s^t)$ and $y(s^{t-1})$. As shown in Fig. 6, we plot the value of comp(s^t)

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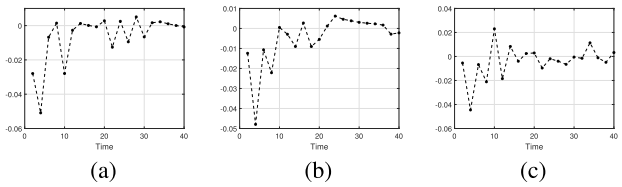


Fig. 6. Complementary information for newly added scale at time t, namely $\text{comp}(s^t)$ is illustrated in (a) Indian Pines data set, (b) Salinas data set, and (c) University of Pavia data set.

Algorithm 1 Proposed AMS Algorithm

Input: A hyperspectral image, I; the corresponding number of classes, C.

Output: A set of multiple segmentation scales.

1. Construct the intra–interscale discrimination index by Equation (5);
2. Define the candidate scale pool by Equation (1);
3. Decide the initial scale in candidates by Equation (6);
4. Adaptively choose the selected scales at time $t+1$ by Equation (7);
5. Determine the final scales when no evidently complementary information is added by Equation (9).

according to time t. We found that with the increase of t, the value $\text{comp}(s^t)$ tends to be zero and will be stable, which means that the newly added scale would provide less complementary information but more computational complexity. Thus, a set of selected scales can be finally determined on the basis of a suitable inflection of the variation tendency

$$\begin{aligned} S^{\text{final}} &= \{s^1, s^2, \ldots, s^t\} \\ &\text{s.t. } \text{comp}(s^t) > 0 \\ &(\text{comp}(s^t) - \text{comp}(s^{t-1})) < 0. \end{aligned} \tag{9}$$

Subsequently, S^{final} is treated as the input of the multiscale fusion framework, as shown in Fig. 2. The spectral–spatial features in different scales are extracted for the classifiers. Then, the final classification result is generated by majority voting, which is utilized in all classification maps pixel by pixel. The pseudocodes of the proposed AMS can be fully described in Algorithm 1.

IV. Experimental Results and Discussion

In this section, we first introduce three standard HSI data sets and the experimental settings utilized in our experiments. Then, the classification results of the proposed AMS are compared with various baselines. Moreover, some deep analysis is presented to further demonstrate the effectiveness of our method.

A. Data Sets

1) Indian Pines: The Indian Pines image was collected by the AVIRIS sensor over an agricultural field in northwestern Indiana. The scene consists of two-thirds agriculture, and one-third forest, and others are natural perennial vegetation. The pixels on the Indian Pines data set are 145 × 145 with 20 m per pixel and 220 bands in the wavelength range of 0.4–2.45 m. This article reserves 200 bands after removing 24 water absorption bands, which is a universally adopted strategy [16], [44], [45] for HSI classification. The groundtruth map utilizes 10 249 labeled pixels with 16 different surface classes.

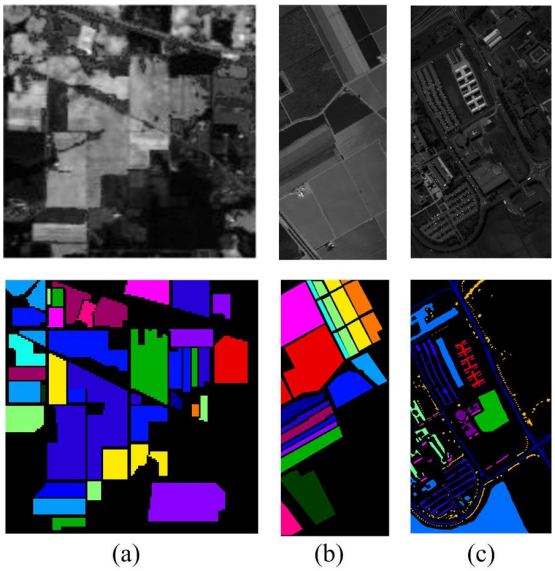


Fig. 7. Gray images (first row) and the groundtruth reference image (second row) of three HSI data sets. (a) Indian Pines. (b) Salinas Scene. (c) University of Pavia.

2) Salinas: The Salinas scene was captured by the 224-band AVIRIS sensor over Salinas Valley, CA, USA. It generates 512 × 217 pixels over 0.4–2.5 μm with a spatial resolution of 3.7 m. Hyperspectral data with 204 bands are adopted after discarding 20 water absorption bands. In this scene, the groundtruth map contains 54 129 labeled pixels with 16 classes.

3) University of Pavia: This data set includes a hyperspectral image of 610 × 340 pixels with 1.3 m and 115 spectral bands in the range 0.43–0.86 μm and is acquired by the ROSIS sensor under the HySens project managed by the German Aerospace Agency. In the groundtruth map, 42 776 labeled pixels with nine classes are adopted and 12 labeled pixels with noise and water bands are removed.

The samples of different land cover classes and the corresponding false colors are listed in Table I, whereas the gray images and groundtruth reference images are shown in Fig. 7, for Indian Pines, Salinas, and University of Pavia.

B. Experimental Settings

Two kinds of baselines, i.e., popular segmentation-based methods and other state of the art, are considered for evaluating the effectiveness of the proposed AMS. In all experiments, three commonly used metrics, i.e., OA, average accuracy (AA), and Kappa coefficient, are used to measure the performance. To obtain more accurate results, all the experiments are repeated ten times, and the average classification performance is finally reported.

1) Experimental Settings for Comparing With Segmentation-Based Baselines: Following the setting of [23], a total

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TABLE I

NUMBER OF SAMPLES AND THE BACKGROUND COLOR OF LAND COVER CLASSES ON INDIAN PINES, SALINAS SCENE, AND UNIVERSITY OF PAVIA IMAGES

Indian Pines		Salinas Scene		University of Pavia	
Class Names	Numbers	Class Names	Numbers	Class Names	Numbers
Alfalfa	46	Brocoli_green_weeds_1	2009	Asphalt	6631
Corn-notill	1428	Brocoli_green_weeds_2	3726	Bare soil	18649
Corn-mintill	830	Fallow	1976	Bitumen	2099
Corn	237	Fallow_rough_plow	1394	Bricks	3064
Grass-pasture	483	Gravel	1345	Fallow_smooth	2678
Grass-trees	730	Stubble	3959	Meadows	5029
Grass-pasture-mowed	28	Celery	3579	Metal sheets	1330
Hay-windrowed	478	Grapes_untrained	11271	Shadows	3682
Oats	20	Soil_vinyard_develop	6203	Trees	947
Soybean-notill	972	Corn_senesced_green_weeds	3278		
Soybean-mintill	2455	Lettuce_romaine_4wk	1068		
Soybean-clean	593	Lettuce_romaine_5wk	1927		
Wheat	205	Lettuce_romaine_6wk	916		
Woods	1265	Lettuce_romaine_7wk	1070		
Buildings-Grass-Trees-Drives	386	Vinyard_untrained	7268		
Stone-Steel-Towers	93	Vinyard_vertical_trellis	1807		
Total Number	10249	Total Number	54129	Total Number	42776

of 30 training samples are randomly selected for each class to construct the training set, whereas the rest is used to build the testing set on the Indian Pines, Salinas, and University of Pavia data sets. For some particular classes with only a few labeled pixels, e.g., oats and grass-pasture-mowed on the Indian Pines, half of the samples are selected for training, whereas others are used for testing. A general multiscale fusion framework (see Fig. 2) is adopted for all the comparisons to objectively evaluate performance; that is to say, AMS and baselines are applied as input to the same fusion procedure to provide the selected scales. The classification models adopt three typical classifiers, namely, SVM, random forest (RF) [46], [47], and extreme learning machine (ELM) [48], [49], to exclude the interference of different classifiers. Due to the absence of adaptive methods, the popular single-scale and multiscale segmentation-based methods are listed as follows.

1) *Single Scale Based on Texture Ratio (SSTR):* The texture ratio R of HSI is calculated by a certain edge detector, then the scale is determined by the product of fixed basic superpixel number S^{base} and R. Following the setting of [22], S^{base} is set to 2000 on all data sets, and the texture detector employs LoG.
2) *Ideal Single Scale (ISS):* Assuming that classification performance at each scale is known, it can select the ideally optimal single scale with the best OA. Concretely, ISS are 98, 81, and 29 for the Indian Pines, Salinas Scene, and University of Pavia, respectively.
3) *Manual Multiple Scales (MMSs):* A commonly used strategy in many studies is to segment the first principle component of input HSI into $2C + 1$ scales, and the number of superpixel regions in cth scale is $s_c = (\sqrt{2})^c \times s_f$, $c = 0, \pm 1, \pm 2, \ldots, \pm C$. By referring to the setting of [23], we select the value of C with the best performance, $C \in [5, 6, 7, 8]$. s_f is set to 100, 100, and 20 for Indian Pines, Salinas, and University of Pavia, respectively.
4) *Best N Multiple Scales (BNMSs):* Assuming that the scales with top-t OAs are known, then the fusion results of these scales can be tested. The value of t refers to (9) of AMS for equal comparisons.

Compared with the single-scale baselines (i.e., SSTR and ISS), our adaptive single scale (ASS) shows that the best intra–interscale discrimination index is adaptively selected by (6) is evaluated. Compared with the multiscale fusion baselines, our AMS in the candidate scale pool S is tested.

2) Experimental Settings for Comparing With Other State of the Art: Deep learning based method is a recent hotspot [50] and always shows remarkable performance in HSI classification community [51]–[55]. Thus, our AMS based on SVM is compared with several deep learning based state of the art, including CNN-PPFs [56], Gabor-CNN [57], S-CNN [58], 3-D-CNN [15], 3-D-GAN [45], and DFFN [59]. Following the setting of [51], a total of 200 training samples from each class are randomly chosen for training, whereas the rest is adopted for testing. Some of the classes on the Indian Pines are with a small number of labeled pixels, which are insufficient for training robust deep learning models. So Salinas and University of Pavia data sets are utilized in this experiment.

C. Comparisons With Segmentation-Based Baselines

ASS, as the scale with the best intra–interscale discrimination index, is compared with ISS and SSTR, which also pertains to single-scale selection. As shown in Tables II–IV, three phenomena can be observed. First, ISS represents the limitation of the selection of the single scale, the performance of which is always ideally optimal for the single-scale HSI classification. Second, by benefiting from available texture information, the results of SSTR seem to perform well on Indian Pines and Salinas data sets. However, the OA, AA, and Kappa of SSTR on the University of Pavia data set are somewhat unsatisfactory, which means that finding a fixed S^{base} adaptable to various data sets is difficult. Third, compared with ISS and SSTR, the selected scales of ASS over

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TABLE II
COMPARISONS WITH SEGMENTATION-BASED BASELINES OVER THREE TYPICAL CLASSIFIERS ON INDIAN PINES DATA SET

Classifiers	Metrics	Single Scale			Multi-scale Fusion		
		SSTR	ISS	ASS	MMS	BNMS	AMS
SVM	OA	0.8865	0.9409	0.8965	0.9560	0.9522	**0.9651**
	AA	0.9471	0.9654	0.9218	0.9709	0.9709	**0.9764**
	Kappa	0.8708	0.9325	0.8816	0.9496	0.9478	**0.9602**
RF	OA	0.8957	0.9403	0.8784	0.9585	0.9483	**0.9619**
	AA	0.9467	0.9627	0.8945	0.9724	0.9672	**0.9754**
	Kappa	0.8811	0.9316	0.8613	0.9525	0.9401	**0.9566**
ELM	OA	0.9093	0.9485	0.9182	0.9570	0.9567	**0.9646**
	AA	0.9547	0.9654	0.9400	0.9723	0.9733	**0.9802**
	Kappa	0.8967	0.9314	0.9064	0.9484	0.9501	**0.9597**

TABLE III
COMPARISONS WITH SEGMENTATION-BASED BASELINES OVER THREE TYPICAL CLASSIFIERS ON SALINAS DATA SET

Classifiers	Metrics	Single Scale			Multi-scale Fusion		
		SSTR	ISS	ASS	MMS	BNMS	AMS
SVM	OA	0.9498	**0.9935**	0.9678	0.9885	0.9898	0.9919
	AA	0.9476	**0.9931**	0.9683	0.9889	0.9901	0.9921
	Kappa	0.9442	**0.9927**	0.9641	0.9871	0.9875	0.9917
RF	OA	0.9658	**0.9925**	0.9703	0.9881	0.9811	0.9921
	AA	0.9743	0.9885	0.9733	0.9896	0.9865	**0.9923**
	Kappa	0.9620	0.9917	0.9669	0.9868	0.9789	**0.9919**
ELM	OA	0.9635	0.9927	0.9697	0.9888	0.9881	**0.9928**
	AA	0.9757	0.9903	0.9695	0.9901	0.9903	**0.9916**
	Kappa	0.9594	**0.9919**	0.9662	0.9868	0.9871	0.9918

TABLE IV
COMPARISONS WITH SEGMENTATION-BASED BASELINES OVER THREE TYPICAL CLASSIFIERS ON UNIVERSITY OF PAVIA DATA SET

Classifiers	Metrics	Single Scale			Multi-scale Fusion		
		SSTR	ISS	ASS	MMS	BNMS	AMS
SVM	OA	0.6049	0.8698	0.8126	0.9086	0.9363	**0.9519**
	AA	0.7203	0.9031	0.8612	0.9285	0.9429	**0.9558**
	Kappa	0.5236	0.8297	0.7588	0.8805	0.9167	**0.9364**
RF	OA	0.7480	0.8975	0.8940	0.9196	0.9358	**0.9415**
	AA	0.8264	0.9115	0.9071	0.9427	0.9521	**0.9548**
	Kappa	0.6848	0.8650	0.8602	0.8948	0.9157	**0.9232**
ELM	OA	0.7766	0.8960	0.8894	0.9324	0.9456	**0.9488**
	AA	0.7571	0.8444	0.8441	0.9077	0.9116	**0.9182**
	Kappa	0.7175	0.8504	0.8497	0.9111	0.9301	**0.9313**

SVM, RF, and ELM are all good on three data sets. Although SSTR has a better performance than ASS on Indian Pines, its stability is significantly worse. Above all, the intra–interscale discrimination index is proved to be effectively applied to adaptively seek a robust scale for HSI classification.

Tables II–IV also show the results of multiscale fusion methods, the performance of which is usually better than that of the single scale methods, especially on the Indian Pines and University of Pavia. However, some of the results of ISS are the best on Salinas. Despite that the best single scale is actually unknown without a manual setting; a possible reason can be concluded that the distribution of land cover on Salinas is obviously simpler than that on the other two data sets. The performance of SSTR that is poor on the University of Pavia is even close to those of other baselines on Salinas. Thus, classification results based on the best single scale may be competitive in such a simple HSI scene. Compared with the MMS and BNMS, the OA, AA, and Kappa of AMS all surpass those of multiscale fusion baselines over three classifiers. This finding demonstrates that the multiple scales adaptively obtained from AMS perform well and are even better than those obtained from manual strategies. Moreover, complementary information is useful for choosing the suitable scales, and the fusion of scales that having optimal performance does not always lead to better classification result.

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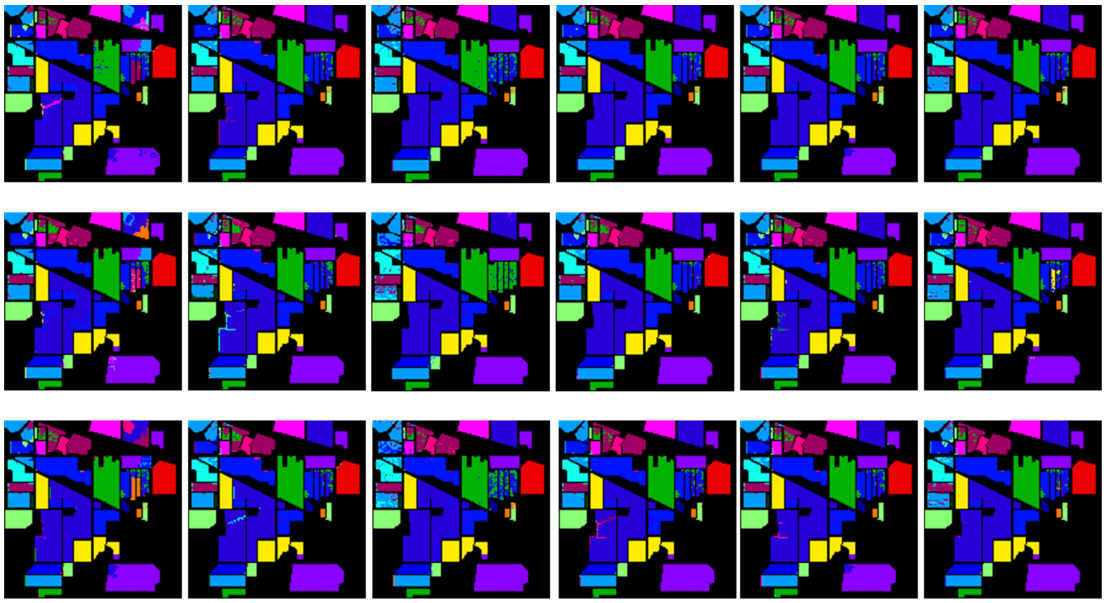


Fig. 8. Classification maps of different methods (each column represents different methods) over three classifiers (each row represents different classifiers) on Indian Pines data set. From the first row to the last row: SVM, RF, and ELM. (a) SSTR. (b) ISS. (c) ASS. (d) MMS. (e) BNMS. (f) AMS.

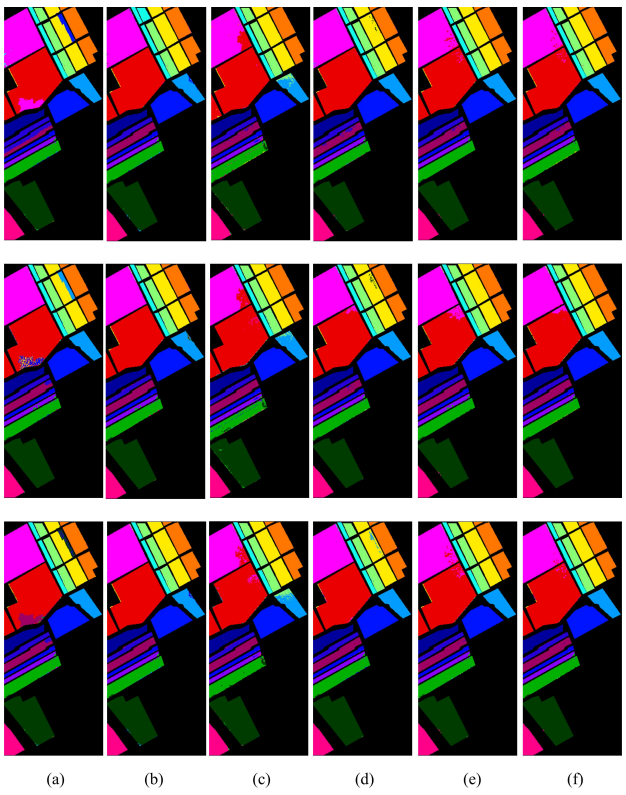


Fig. 9. Classification maps of different methods (each column represents different methods) over three classifiers (each row represents different classifiers) on Salinas data set. From the first row to the last row: SVM, RF, and ELM.(a) SSTR. (b) ISS. (c) ASS. (d) MMS. (e) BNMS. (f) AMS.

The visual performance comparison of different methods over SVM, RF, and ELM are shown in Figs. 8–10, on the Indian Pines, Salinas, and University of PaviaU data sets, respectively. The proposed AMS can provide robust visual classification results compared with other methods. To further explain the robustness of AMS, more results are reported in two aspects.

1) Fusion details, including OAs of the best scale and the worst scale before multiscale fusion, are compared with the final fusion results over SVM. As shown in Fig. 11, the results of AMS are always better than those before multiscale fusion, but the performance of BNMS is lower compared with that of the best scale on Salinas. Second, compared with MMS and BNMS, the AMS adopts more scales with bad performance. By contrast, our improvements by using the multiscale fusion are even greater. Third, BNMS has a better development on the University of Pavia than itself on the other two data sets, and the performance difference between the best scale and the worst scale on the University of Pavia is also larger. The scales with obvious performance differences are likely to result in greater improvement in multiscale fusion.
2) The fusion results of random scales in the candidate scale pool S for 200 iterations are evaluated over SVM. As shown in Fig. 12, the OAs based on random scales fusion (blue imaginary line) can drastically fluctuate. By contrast, our performances (red lines) are always at the top part of the OA curves on all three data sets, proving that the choice of multiple scales in AMS is reliable and satisfactory.

D. Comparisons With State of the Arts

This section presents the performance comparisons of our AMS over SVM classifier and state of the art on the Salinas and University of Pavia data sets, as shown in Tables V and VI. First, the results of AMS obviously exceeds than those of original SVM, e.g., 8.82% and 8.05% OA improvements exist

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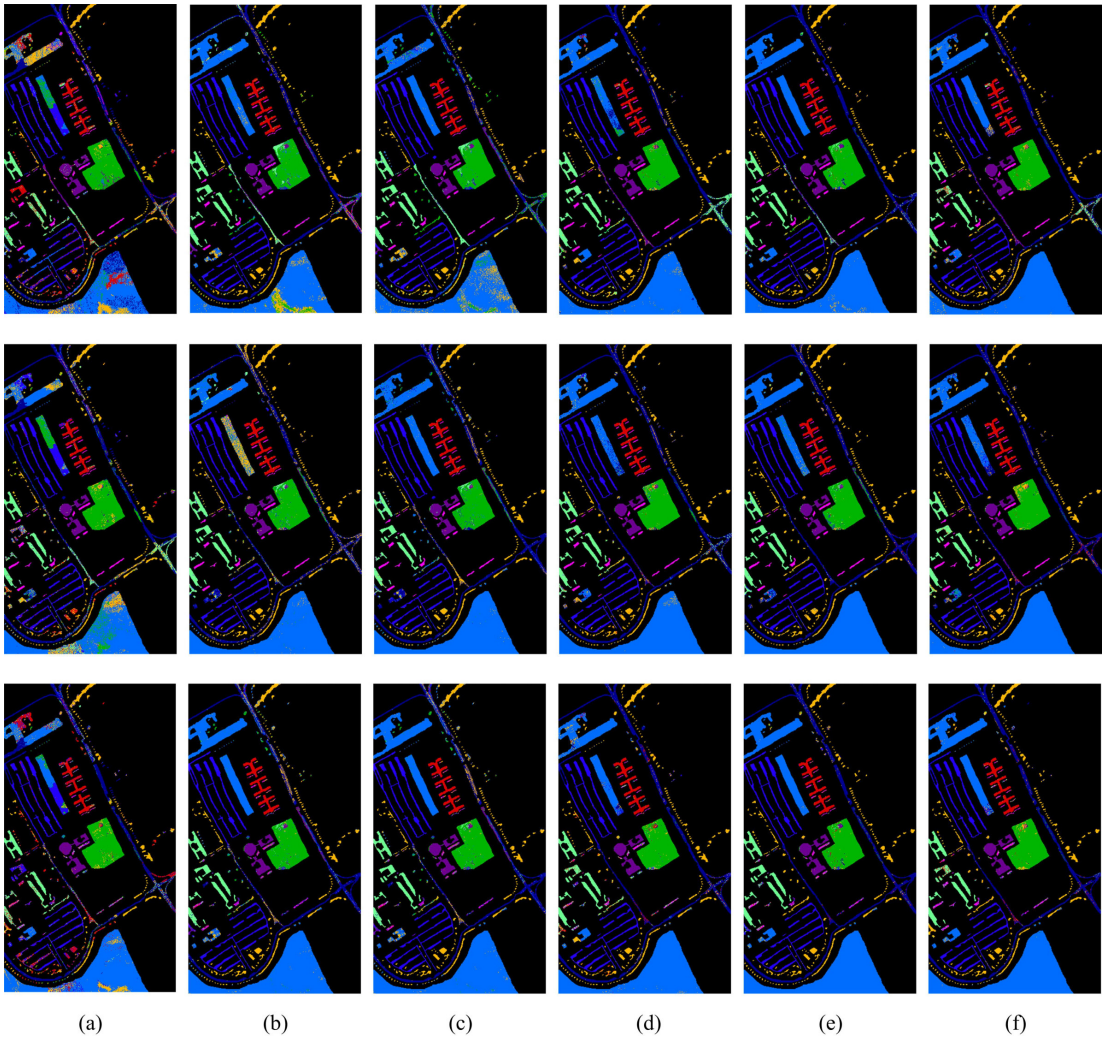


Fig. 10. Classification maps of different methods (each column represents different methods) over three classifiers (each row represents different classifiers) on the University of PaviaU data set. From the first row to the last row: SVM, RF, and ELM. (a) SSTR. (b) ISS. (c) ASS. (d) MMS. (e) BNMS. (f) AMS.

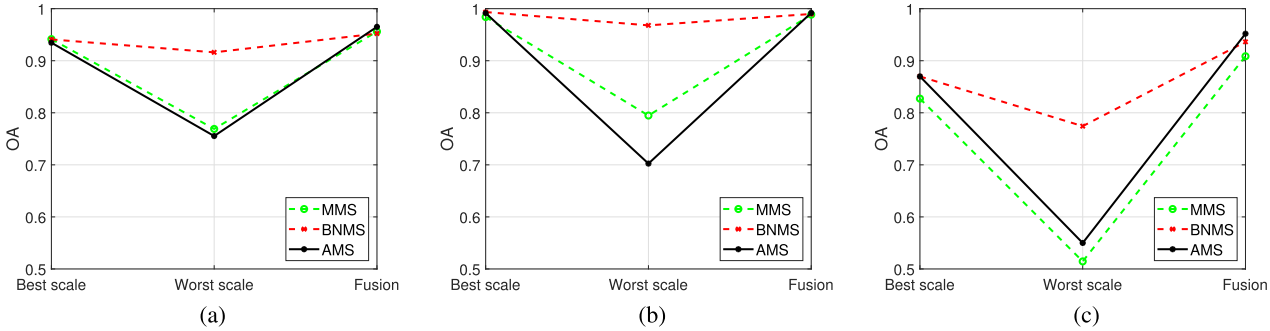


Fig. 11. Best scale, worst scale, and fusion results of the MMS, BNMS, and AMS are labeled in red "x," green "o," and black "*" on (a) Indian Pines data set, (b) Salinas data set, and (c) University of Pavia data set.

on the Salinas and University of Pavia data sets, respectively. The proposed framework is remarkable for developing the performance of classifiers. Second, the OA and Kappa of AMS are the optimal compared with state of the arts on two data sets. Although the AA of DFFN is a little higher than that of AMS, our performance is still very close to DFFN and evidently surpasses all other baselines. The robustness of the proposed method is further proved. Third, the class accuracies of most baselines are unstable, such as class 8 of CNN-PPFs and class 2 of Gabor-CNN on Salinas, class 3 of 3-D-CNN, and class 7 of 3-D-GAN on the University of Pavia. By contrast, our superior class accuracies have demonstrated the stability of AMS.

E. Influence of Different k in Candidate Scale Pool

The candidate scale pool S is set as $[S^{\text{lower}}:k:S^{\text{upper}}]$ in this article, where k is the sole parameter in AMS and can be used to reduce the computational complexity of multiple scales selection. Thus, this section evaluates whether different

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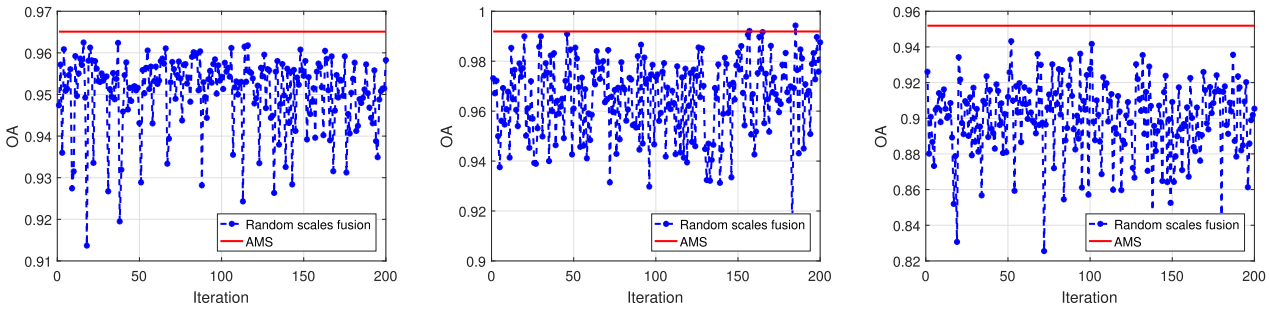


Fig. 12. Results of comparison of the fusions of random scales and AMS are labeled in blue imaginary line and red line, on (a) Indian Pines data set, (b) Salinas data set, and (c) University of Pavia data set, respectively.

TABLE V

COMPARISONS OF AMS AND STATE OF THE ARTS ON SALINAS DATA SET

Classes	SVM	CNN-PPFs	Gabor-CNN	S-CNN	3D-CNN	3D-GAN	DFFN	AMS
1	0.9961	0.9984	**1.0000**	**1.0000**	0.9994	0.7535	0.9999	**1.0000**
2	0.9969	0.9977	0.8806	0.9775	0.8545	0.9849	0.9994	**1.0000**
3	0.9956	0.9811	0.9925	0.9888	**1.0000**	**1.0000**	**1.0000**	**1.0000**
4	0.9941	0.9957	**1.0000**	**1.0000**	0.9977	0.9976	**1.0000**	**1.0000**
5	0.9872	0.9854	0.9888	0.9993	0.9996	**1.0000**	0.9911	0.9963
6	0.9977	0.9992	**1.0000**	0.8948	**1.0000**	0.9997	0.9995	0.9994
7	0.9952	0.9996	0.9894	0.9930	0.9960	0.9945	0.9943	**1.0000**
8	0.7674	0.8911	0.9948	0.9869	0.9931	0.9830	0.9956	**0.9993**
9	0.9937	0.9969	0.9727	0.9934	0.9997	0.9995	**1.0000**	0.9996
10	0.9513	0.9778	0.9921	**1.0000**	0.9941	0.9979	0.9979	0.9794
11	0.9932	0.9933	**1.0000**	0.9993	**1.0000**	**1.0000**	0.9948	**1.0000**
12	0.9970	**1.0000**	**1.0000**	0.9989	**1.0000**	**1.0000**	0.9984	**1.0000**
13	0.9915	0.9967	**1.0000**	0.8862	**1.0000**	**1.0000**	0.9996	0.9891
14	0.9838	0.9875	0.9979	0.8872	**1.0000**	**1.0000**	0.9995	**1.0000**
15	0.7556	0.8999	0.9431	0.9062	0.9036	0.9952	0.9945	**0.9976**
16	0.9923	0.9907	0.9338	**0.9996**	0.8593	0.9025	**0.9996**	0.9876
OA	0.9092	0.9487	0.9763	0.9762	0.9728	0.9822	0.9971	**0.9974**
AA	0.9618	0.9807	0.9804	0.9694	0.9748	0.9775	**0.9978**	0.9968
Kappa	0.8985	0.9404	0.9734	0.9510	0.9695	0.9793	0.9967	**0.9971**

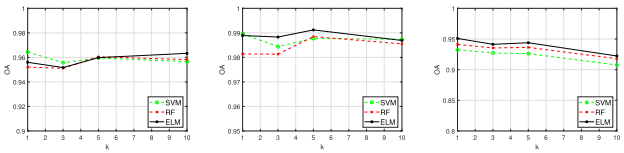
TABLE VI

COMPARISONS OF AMS AND STATE OF THE ARTS ON UNIVERSITY OF PAVIA DATA SET

Classes	SVM	CNN-PPFs	Gabor-CNN	S-CNN	3D-CNN	3D-GAN	DFFN	AMS
1	0.8549	0.9723	**0.9953**	0.9547	0.9903	0.9918	**0.9953**	0.9755
2	0.9212	0.9527	0.9821	0.9871	0.9811	0.9886	0.9771	**0.9933**
3	0.8577	0.9513	0.8974	0.9732	0.8856	0.9494	**0.9989**	0.9905
4	0.9641	0.9689	0.9302	0.9772	0.8351	0.9015	0.9788	**0.9845**
5	0.9860	0.9999	0.9942	**1.0000**	0.9949	0.9949	0.9948	0.9851
6	0.9252	0.9855	0.9877	0.9767	0.9533	0.9856	**0.9969**	0.9881
7	0.9379	0.9656	0.9882	0.9836	0.9631	0.9274	**1.0000**	0.9774
8	0.8656	0.9443	0.9412	0.9565	0.9758	0.9718	0.9859	**0.9925**
9	0.9797	0.9939	0.9791	**1.0000**	0.9625	0.9851	0.9961	0.9921
OA	0.9078	0.9763	0.9733	0.9793	0.9637	0.9781	0.9857	**0.9883**
AA	0.9214	0.9704	0.9662	0.9788	0.9482	0.9665	**0.9916**	0.9866
Kappa	0.8813	0.9690	0.9662	0.9743	0.9502	0.9697	0.9808	**0.9845**

k affects the classification performance of AMS. Specifically, we set $k = 1, 3, 5, 10$ to generate four groups of candidate scales, and then the AMS based on different values of k is tested over SVM, RF, and ELM. As shown in Fig. 13, several observations can be drawn. First, an ascending or flat tendency is more likely observable on Indian Pines and Salinas, whereas a descending tendency exists on the University of Paiva. Second, the maximal differences of OAs over SVM are 1.07%,

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(a) (b) (c)

Fig. 13. OAs of the AMS with different values of k over SVM, RF, and ELM are labeled in green "o," red "x," and black "*," respectively, on (a) Indian Pines data set, (b) Salinas data set, and (c) University of Pavia data set.

TABLE VII
RUNTIME RESULT (S) COMPARISONS ON THREE DATA SETS

Dataset	T	SSTR	MMS	AMS	
				Online	Offline
Indian Pines	30	2.70	19.45	18.92	29.53
	100	4.21	44.07	29.40	29.53
	200	5.70	68.93	43.25	29.53
Salinas	30	9.88	89.63	60.24	257.05
	100	12.18	122.35	85.67	257.05
	200	14.36	163.70	119.47	257.05
University of Pavia	30	8.39	60.65	62.94	680.47
	100	11.84	116.05	93.06	680.47
	200	17.74	203.37	142.17	680.47

0.52%, and 2.45%, those of OAs over RF are 0.94%, 0.72%, and 2.32%, and those of OAs over ELM are 1.16%, 0.43%, and 2.86%, on the Indian Pines, Salinas, and University of Paiva, respectively. Third, compared with the results of $k = 1, 3, 5$, the performance variation is more dramatic when k is 10. In summary, the OAs seem to have no uniform tendency when the value of k is increased, and the performance variation between different k is also minimal. Thus, the performance of AMS is relatively insensitive to the selection of k. Moreover, we still recommend that $k < 10$ because there are few candidate scales when the value of k is much larger, and the corresponding performance is probably unstable.

F. Runtime Analysis

The intra–inter discrimination index and candidate scale pool can be obtained as soon as the HSI is provided. Therefore, the implementation of AMS can be divided into two separate phrases. In the offline part, the intra–inter discrimination index for each scale is computed. In the online part, the processes of iterative multiple scales determination and multiscale fusion are conducted. With the above two-phrase implementation manner, the computational complexity of AMS can be greatly reduced and is mainly influenced by the number and the size of the selected scales. In Table VII, we report the runtime results on the Indian Pines, Salinas, and University of Pavia data sets with different training numbers ($T = 30$, 100 and 200). SSTR, MMS and our AMS over SVM classifier are tested on MATLAB R2018b using an Intel Core i7 CPU with 3.40 GHZ and 16-GB memory personal computer with windows platform. The testing time of all methods is calculated by a single-threaded MATLAB process. The SSTR is a single-scale method and has the lowest runtime, the MMS uses a fixed number of scales and the AMS adopts adaptive number of scales. Despite that the offline runtime of AMS does not involve T, the runtime of all methods would evidently increase along with the growth of training numbers; however, our runtime is still competitive.

V. CONCLUSION

This article proposes a novel AMS method to adaptively determine a set of complementary scales for HSI classification. The proposed framework includes three stages: definition of candidates scale pool, calculation of scale discrimination index, and determination of adaptive multiple scales. The eventual multiple scales are iteratively chosen until no evidently complementary information can be provided by newly added ones. Extensive experimental results on the three HSI data sets have demonstrated the robustness of the proposed AMS. The three conclusions can be drawn as follows. First, the scale with the best intra–interscale discrimination index always accomplishes satisfactory performance, and it can be used to select an appropriate single scale. Second, the AMS can adaptively determine a set of reliable multiple scales and perform well when compared with state-of-the-arts. Third, the parameter k is adopted to reduce the computing complexity of AMS, and the classification performance has been verified as relatively insensitive to k.

In the proposed framework, although seeking an effective initial scale on the basis of intra–inter discrimination index is feasible, the intra–inter discrimination index has no strict correspondence to the classification result. It is hard to choose effective and complementary scales in AMS. Thus, how to construct much better scale features is one of our future directions. Moreover, the class number C is a low boundary for the number of superpixels, the candidate scale pool can be further improved to be more realistic.

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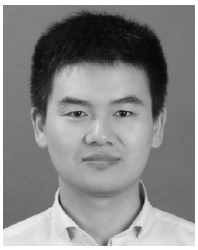
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